

$$Q1) \quad F_{x_1} = F_1 \sin 70^\circ = 350 \sin 70^\circ = 328.9 \text{ N}$$

$$F_{y_1} = F_1 \cos 70^\circ = 350 \cos 70^\circ = 119.71 \text{ N}$$

$$F_{3x} = -180 \times \frac{12}{13} = -166.154 \text{ N}$$

$$F_{3y} = 180 \times \frac{5}{13} = 69.23 \text{ N}$$

$$F_{Rx} = 328.9 - 166.154 = 162.75 \text{ N}$$

$$F_{Ry} = 119.71 + 69.23 + 200 = 388.94 \text{ N}$$

$$A) \quad F_R = \sqrt{162.75^2 + 388.94^2} = 421.62 \text{ N}$$

$$\boxed{F_R = 422 \text{ N}}$$

$$B) \quad \phi = \tan^{-1} \left(\frac{388.94}{162.75} \right) = 67.3^\circ$$

$$\boxed{\phi = 67.3^\circ}$$

$$Q3) \quad \vec{F} = \{300\hat{i} - 100\hat{j} + 400\hat{k}\} \text{ N}$$

coordinates $C(0, 1, 3) \quad A(6, 0, 0)$

$$\vec{CA} = 6\hat{i} - \hat{j} - 3\hat{k}$$

$$\hat{u}_{CA} = \frac{6}{\sqrt{46}}\hat{i} - \frac{1}{\sqrt{46}}\hat{j} - \frac{3}{\sqrt{46}}\hat{k}$$

Projection of $\vec{F}$ along $\vec{CA}$

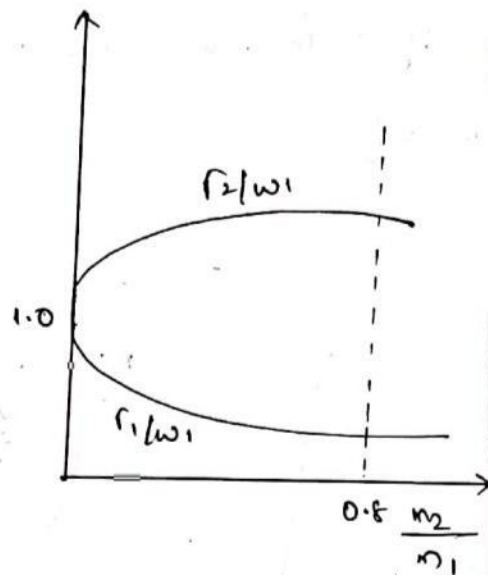
$$\vec{F} \cdot \hat{u}_{CA} = F_{\text{on } CA} = 300 \times \frac{6}{\sqrt{46}} + 100 \times \frac{1}{\sqrt{46}} - 400 \times \frac{3}{\sqrt{46}}$$

$$F_{\text{on } CA} = 103.21 \text{ N} = \boxed{103 \text{ N}}$$

(or)

we can say that for higher range, $\frac{m_2}{m_1} > 0.8$

$$(\omega \neq \omega_1 = \omega_2)$$



$$\checkmark \quad \frac{A_2}{y_0} = \frac{1}{\left[1 + \frac{k_2}{k_1} - \left(\frac{\omega}{\omega_1}\right)^2\right] \left[1 - \left(\frac{\omega}{\omega_2}\right)^2\right] - k_2/k_1}$$

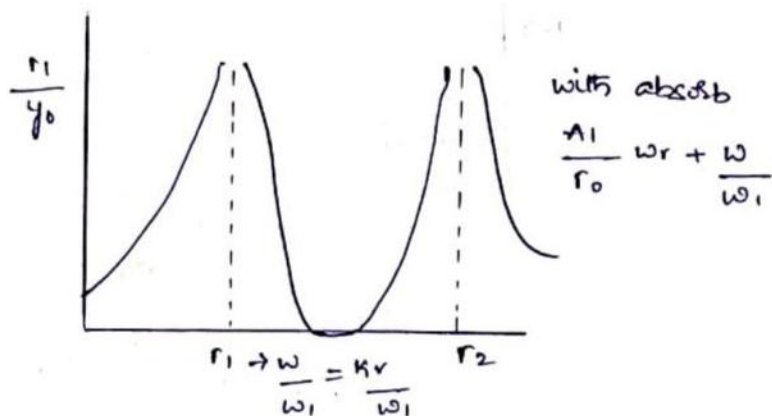
Now $\Rightarrow$ variation of the amplitude of vibration of the

m_1 with cycle speed (v) = ?

$\checkmark$ for any velocity of cycle, $\omega = ?$ we know

$$\boxed{\omega \times v}$$

let $\omega = kv$



$\Rightarrow$ for a constant speed v_0 when corresponding $\omega = \omega_1$

then cycle will be completely vibration free but

also for values of $v_0 \Rightarrow$ vibration will be less

$\Rightarrow$ If we want to increase range at velocity, where vibration will be less than we should take $\frac{m_2}{m_1}$ higher value.

$\Rightarrow$ If $m_2/m_1 \geq 0.8 \Rightarrow$ range will become constant

Problem 4.49)

$$F_1 = \{30\hat{i} + 20\hat{j} - 30\hat{k}\}N, F_2 = \{60\hat{i} - 10\hat{j} - 20\hat{k}\}N$$

$$\vec{r}_{AB} = 1.2\hat{j}$$

$$\vec{r}_{AC} = -0.4\hat{i} + 1.2\hat{j}$$

$$\vec{M}_{1A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1.2 & 0 \\ 30 & 20 & -30 \end{vmatrix} = \{-36\hat{i} - 36\hat{k}\}N.m$$

Part A)

$$\boxed{-36, 0, -36}$$

$$\vec{M}_{2A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -0.4 & 1.2 & 0 \\ 60 & -10 & -20 \end{vmatrix} = -24\hat{i} - 8\hat{j} - 68\hat{k}$$

Part B)

$$\boxed{-24, -8, -68}$$

Given data:-

2 DOF system

Here DOF = Degree of freedom

So,

Consider the 2 degree of freedom vibration system consisting of a $m_1 - c_1 - k_1$ primary system and a $m_2 - c_2 - k_2$ absorber system, see fig below. Given a simple harmonic excitation, the vibration of the absorber system can provide the right amount of force to keep the primary system completely still.

from, the 2 DOF system:-

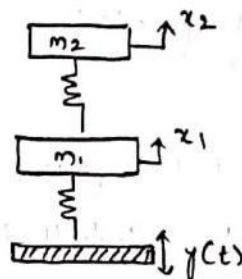
$m_1, c_1, k_1 \Rightarrow$ primary system

$m_2, c_2, k_2 \Rightarrow$ absorber system

$c_1 = c_2 = 0 \Rightarrow m_1, k_1 \text{ \& } m_2, k_2$

Base motion $y(t) = y_0 \sin(\omega t)$

So, from the below diagram



Q4.2) $F = 300 \text{ N}$

A) components of force $F_x = 300 \times \frac{4}{5} = 240 \text{ N}$

$$F_y = -300 \times \frac{3}{5} = -180 \text{ N}$$

$$M_o = (240 \times 1) \text{ Nm} - (180 \times 2) \text{ Nm} = -120 \text{ N.m}$$

$$\boxed{M_o = -120 \text{ N.m}}$$

B) $F_x = 300 \times \frac{4}{5} = 240 \text{ N}$, $F_y = -180 \text{ N}$.

$$M_o = (-240 \times 3) + (180 \times 1) = -540 \text{ N.m}$$

$$\boxed{M_o = -540 \text{ N.m}}$$

Problem 4.32 $\vec{F} = \{45\hat{i} + 54\hat{j} + 27\hat{k}\} \text{ lb}$

$$\vec{AB} = \hat{i} + (2+3\cos 45)\hat{j} + (3\sin 45)\hat{k} = \hat{i} + 4.121\hat{j} + 2.121\hat{k}$$

$$\vec{M} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4.121 & 2.121 \\ 45 & 54 & 27 \end{vmatrix} = -3.267\hat{i} + 68.445\hat{j} - 131.445\hat{k}$$

A) Magnitude $M = \sqrt{3.267^2 + 68.445^2 + 131.445^2} = 148.23 \text{ lb.ft}$

$$\boxed{M = 148 \text{ lb.ft}}$$

B) $\cos \alpha = \frac{-3.267}{148.23} \Rightarrow \alpha = 91.3^\circ$

$$\cos \beta = \frac{68.445}{148.23} \Rightarrow \beta = 62.5^\circ$$

$$\cos \gamma = \frac{-131.445}{148.23} \Rightarrow \gamma = 152.5^\circ$$

$$\boxed{(91.3, 62.5, 152)}$$